Using the Lab Grading Server

Pre-Requisites:

1. Run the SCS account management tool to update your account and be added to the course OpenStack project: https://carleton.ca/scs/tech-support/accounts/#manage-scs-account
2. Create an OpenStack instance (only one instance is required per group):
    a. Use the COMP4601A-W26.2026-01-08 instance snapshot
    b. Use the default flavor
    c. Add the ping-ssh-egress security group to support SSH
    d. Add the web3000 security group to open port 3000 for your server
3. Assign a floating IP to your instance
4. Connect to and install whatever software you need on the instance
5. Find your server key under the “Lab Grading Server Key” entry in the grades section on Brightspace
6. You will have to be connected to the Carleton VPN or present on campus to connect to OpenStack.

Registering Your Server:

Make a POST request to http://134.117.26.91:3000/servers. Ensure the Content-Type header is set to “application/json”. The body should be formatted as follows (all entries are strings, including student IDs):

{
“serverAddress” : “http://***yourInstanceFloatingIP***:3000”,
“members” : [***studentId#1, studentId#2 (if applicable), studentId#3 (if applicable)***],
“key” : “***yourServerKey-can-be-any-one-of-member’s-keys***”
}

If done correctly, you should receive a response with status code 201 and a body containing your server’s name. Record your server’s name somewhere, as you will need it later. Your server’s name should now show up in the list of lab results: http://134.117.26.91:3000/results

Lab Results Page:

A summary of lab results is available at http://134.117.26.91:3000/results. This will show all registered servers and their success/failure for each lab/assignment. Each server/lab combination will have two symbols (✓ or ✗) indicating whether the server has ever passed the tests for that lab and whether the server has passed any of the most recent tests for that lab. Clicking the ? beside the symbols for the lab will take you to a page summarizing the recent tests. If any were failed, a basic error message will be included that may help determine the problem.

Currently, the plan is to assign the auto-graded lab grades based on whether the server ever passed the test before the lab deadline, so you can stop supporting past lab functionality once you have completed the lab.

Updating Lab/Assignment Status:
The central server will only execute tests for labs/assignments that you indicate are running. This way you can disable labs once you have finished them and avoid unnecessary requests coming in to your server. To update the running status of your server, make a PUT request to http://134.117.26.91:3000/servers/***yourServerName***. An example of what format the body of the request should have is below:

```
{
  "key" : "yourServerKey",
  "running" : {
    "INFO" : true,
    "L3" : true,
    "L4" : false,
    "A1" : false
  }
}
```

Each lab/assignment will have a name, which you can see in the lab results list (L3, L4, A1, A2, etc.). There is also an "INFO" test that is the equivalent to a 'hello world' test for your server (details below). Any keys not included in the "running" map will default to false. You can see the running status of your own server by going to http://134.117.26.91:3000/servers/***yourServerName***/running or by clicking the ? beside your server name in the lab results list.

The "INFO" Test:
This is ungraded but will give you a way to ensure you have your server registered properly. To pass this test, your server must respond to GET requests for /info with JSON data indicating the name of the server: {"name" : "**yourServerName**"}.